Moving the sun in architecture photos

Mehmet Ozturk | github.com/mehmetztrk/sun-direction-relighting

What I did

I tried to change where the sun comes from in a photo without changing the building. I took three real photos from Unsplash (a villa, a concrete block and a living room) and tried three AI models on them. I also wrote one small script of my own.

Results in one look

- **Sun-Direction LoRA:** the best one. It really moves the sun on exteriors, but it changes some materials.
- **Qwen 3D Lighting:** fast and keeps the colors, but it zooms the photo and old shadows stay.
- **FLUX.1 Kontext:** keeps the building, but the sun doesn't move on exteriors.
- **My light transfer script:** furniture stays perfect, but the light barely changes.
- **Interiors:** nothing worked well enough yet.

Time and cost

- Sun-Direction: 20 to 26 s on a free Hugging Face Space, 73 to 116 s on my RTX 4070
- Qwen 3D Lighting: 6.5 to 17 s on a free Space
- FLUX.1 Kontext: 31 to 45 s on a free Space
- Everything cost me $0. Space times include the queue.

What I would ship

The Sun-Direction 4B model on a serverless GPU, for exteriors only, with an automatic check that retries if the building changes. My estimate is about $0.01 to $0.02 per image and 20 to 26 s. That estimate uses RunPod prices and my Space times. I did not test it on that GPU.

1. Sun-Direction LoRA

What I did: this model is made for moving the sun. You give it a small picture of a lit ball and it copies that light. I first ran it on a free Space, then on my own GPU with the smaller 4B version, because 4B has a license that allows commercial use. I also added a color prompt, and for the villa a stronger sun.



Villa, original



Sun behind the camera
Even light on the front.



Sun behind the building
The sun shows in the sky and walls go into shade.



Sun straight above
Short, hard shadows under the overhangs.



Sunset on the left
The biggest change.



Low sun from the left
Long shadows across the terrace.



Low sun from the right
Left and right are the closest pair.



Concrete block, original
Sun comes from the left.



Sun moved to the right
Blue sky, paler concrete. It added a tree.



Living room, original



Low sun from the back left
Good light, but rug, armchair and window view changed.

Result

Good on exteriors. You can clearly see where the sun is. Not good enough on interiors, and the model was not trained on them.

Problems

- Colors turned gray until I added a color prompt.
- The steel sculpture looks like stone, and the pool loses color.
- It adds things like a tree and clouds.
- When I asked for a sunset on the right, the orange sky stayed on the left.

2. Qwen 3D Lighting

What I did: I used a free Space. You pick the light direction with an angle.



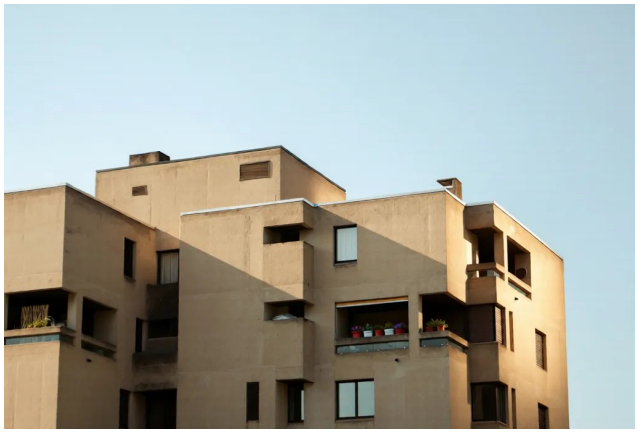
Villa, original



Result
Haze on the left, and the photo got zoomed.



Concrete block, original



Result
Old shadow stayed. It added flower pots.



Living room, original



Result
Furniture stayed. The sun patch on the window wall can't happen.

Result

Fast and keeps colors, but the light doesn't really move.

Problems

- It zooms the photo by 16 to 19 percent every time.
- Old shadows stay, and it adds objects.

3. FLUX.1 Kontext

What I did: I used a free Space and asked in plain text: change only the lighting, keep the building.



Villa, original



Result
Almost nothing changed.



Concrete block, original



Result
The diagonal shadow did not move.



Living room, original



Result
Nice sunset, but all orange, with a new view.

Result

It keeps the building, but it doesn't move the sun on exteriors.

Problems

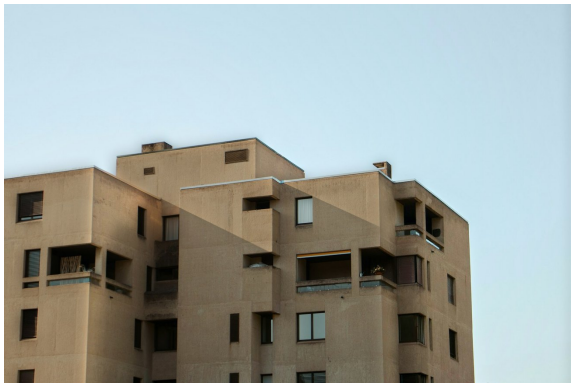
- Inside, the sun appeared on the wrong side and the window view changed.
- The open version is non-commercial.

4. My light transfer script

What I did: the models kept redrawing things, so I kept the original photo and copied only the brightness change from a model's result.



Concrete block, original



Result
The old shadow is still there.



Living room, original



Result
Furniture is safe, and the light change is weak.

Result

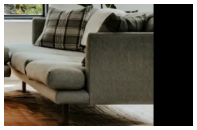
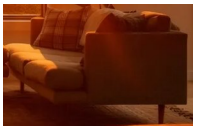
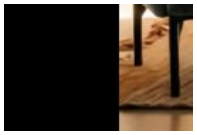
Objects stay perfect, but the sun doesn't really move.

Problems

- Old hard shadows stay.
- Sharper settings add bright halos around objects.

Furniture check

I compared each object up close, and this showed what the overall picture hid. Black areas fell outside that model's frame.

	Original	Sun 9B	Sun 4B	Qwen	Kontext	Transfer
Armchair						
Sofa						
Table						
Rug						

Problems along the way

- Some free Hugging Face demos were paused or crashing, so I found others.
- The free GPU quota ran out quickly, so I moved the Sun-Direction tests to my own computer.
- The bigger 9B model doesn't fit on my 12 GB card, so I used 4B.
- My first score was misleading. Kontext scored best because it barely changed anything, so I checked by eye too.

What can be improved

- Test on at least 30 exteriors, since three photos isn't many.
- Protect glass, metal and water with masks, so they don't change.
- Use a depth map to work out real shadows from the sun position.
- Combine the model's new shadows with the original textures.
- Use masks or a room-trained model to lock furniture in interiors.
- Test on a bigger GPU to get closer to your 13.4 s average.
- Let the user drag the sun, or pick a date, time and location.
- Test on renders too, since I only used real photos.

Other models worth trying

- **Sun-Direction v2:** the author says it's in progress, with no overcast step and control over light hardness and color.
- **Qwen Light Migration:** copies the light from a reference photo, so you could use a real sunset photo.
- **Qwen Edit 2511 with LightingRemap:** you paint color blocks at the edge of the image to set the light. Still an alpha.
- **Nano Banana (Gemini):** paid prompt-based editing from about $0.03 per image.
- **FLUX.2 Edit and Seedream Edit:** newer paid edit models on fal.
- **GPT Image 2:** OpenAI's edit model. I couldn't confirm its price.
- **LTX-2.3 Relight:** sets sun direction for exterior video, so it could also suit a time-of-day feature.
- **HELIOS:** a research project with sun angle control from GPS. I couldn't confirm that its code is public.

Code and all tries, including the failed ones: https://github.com/mehmetztrk/sun-direction-relighting

Photos from Unsplash: framesforyourheart, Clay Banks, Valentin Gerardi.